DIALGA: A Factorized, Camera-Aware Video Embedding with a Spatial Dynamics Code

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Abstract

Video VAEs compress video into a rich but entangled latent: what a scene contains and how it moves are fused into one code. Splitting them normally means training a new VAE. We present DIALGA, a small adapter that re-encodes the latent of a frozen video VAE into two codes — a static code for the scene and a per-frame dynamics code — and we find that the split is won or lost in the decoder, not in the loss. Given a decoder that may read both codes freely, the dynamics code simply absorbs everything: deleting the static code costs only 11% reconstruction, and an independence penalty does not help, because it is satisfied perfectly by a static code carrying noise. We instead make the split structural. The decoder reconstructs each frame as a static image plus a zero-mean per-frame residual, so the time-constant part of the output can only come from the static code. Deleting that code now costs 484%. Freed of redundancy, the same model runs at 24× compression — half the rate of our prior design — where it matches VideoFlexTok and VideoMAE on reconstruction under a matched protocol while carrying a factorization neither expresses. On 163,717 Something-Something-v2 clips the dynamics code reads actions at 10.3% against the static code's 6.1%. We report the costs plainly: absolute fidelity drops 2.3 dB, and semantic accuracy is capped by the frozen latent, not by our encoder.

1 Introduction

A video representation should keep its kinds of information in different places: identity — what the objects are, constant over time — object motion — how they move — and, for a moving camera, camera motion — how the viewpoint moves. Modern video VAEs do the opposite. Trained to reconstruct, they pack texture, identity, object motion, and camera motion into a single unstructured latent grid that offers no axis along which a downstream user can read out identity, freeze dynamics, or subtract the camera (Tong et al., 2022; Bardes et al., 2024). That structure can be recovered, but expensively: sequential VAEs that split content from motion do so by training a VAE from scratch (Zhu et al., 2020; Bai et al., 2021), and recent large predictive video VAEs enrich the latent's motion-awareness but leave it entangled, and cost a full pretraining run (Zhao et al., 2026). We ask a cheaper question: given an already-trained, frozen video VAE, can we recover a named, factorized, camera-aware decomposition of its latent without retraining it?

DIALGA is a lightweight adapter that re-encodes each short chunk of a frozen video-VAE latent into three named, independently accessible slots — a spatial static code (identity), a per-frame spatial dynamics code (object motion), and a camera code — with a decoder structurally prevented from mixing them. Decoding, forward prediction, and counterfactual editing are probes of the factorization, not the objective. We are explicit about the regime: at this extreme compression the code does not beat foundation encoders on semantic accuracy, nor the frozen VAE on reconstruction — it is distilled from that VAE and cannot exceed it, and we report both plainly (Tables 1 and 11). At 288× compression it stays as semantically accessible as the full latent it is distilled from (attribute-probe mAP within noise; Table 1) while remaining decodable (SSIM ≈ 0.96). The contribution is not a higher number on any single axis — it is the decomposition itself: a cheap, camera-aware factorization that entangled latents, whether raw or predictive, cannot provide, and that we show is clean (a diagonal-dominant cross-probe), controllable (identity/motion swap), and transferable (the dynamics code drives robot control).

The line of work that targets this decomposition — sequential VAEs that split video into a time-invariant content code and a per-frame motion code (Li & Mandt, 2018; Zhu et al., 2020; Bai et al., 2021), and object-centric models that factorize into slots (Wu et al., 2023) — validates the factorization almost exclusively on synthetic video with rigid, low-rank motion (Sprites, MUG, CLEVRER). It has remained unclear whether such a factorization can be learned at high compression on real, cluttered, moving-camera video, where the demonstration and the demand are both far harder.

We study DIALGA, an encoder that compresses each short chunk of a frozen video-VAE latent into two named slots — a spatial static code z_{static} and a per-frame dynamics code z_{dyn} — at extreme rate, with a decoder structurally prevented from mixing them. Decoding, forward prediction, and counterfactual editing are probes of the representation, not the objective: the claim is about what the latent carries, not about pixel fidelity, which classical codecs win outright at matched bitrate (Wiegand et al., 2003). Taking this design to a real robot's moving wrist camera (DROID (Khazatsky et al., 2024)) surfaced the failure that became the technical core of this paper. A model that factorizes cleanly on synthetic CLEVRER (Yi et al., 2020) reconstructed real video at only 16 dB — 17 dB below the frozen-VAE ceiling — no matter how the camera was conditioned. The camera was not the bottleneck; reconstruction was.

We trace the failure to a single property shared with the entire prior factorized-video family: the dynamics code is a global per-frame vector, which the decoder can only broadcast uniformly across space. Such a code produces a spatially-uniform, rank-limited per-frame delta field — enough for the near-rigid motion of synthetic scenes, but provably unable to represent the per-pixel, high-frequency temporal change of real textured video. Giving the dynamics code a spatial axis — a small per-frame feature grid rather than a vector, in line with what every real-video tokenizer converges on (Wang et al., 2025; NVIDIA, 2025; Yu et al., 2024) — raises held-out reconstruction on DROID from 16.1 to 20.5 dB. A matched-rate ablation (Table 7) attributes the gain to the spatial structure (+2.7 dB), not to the extra rate the change costs (+0.6 dB). The lesson is concrete and transferable: the static code can be a shared grid, but the dynamics code must be spatial to survive real video. On top of the now-reconstructable representation we add a known-camera-pose conditioning path, so the static code can be pushed toward viewpoint stability under a moving camera — an add-on in the spirit of geometry-aware attention (Miyato et al., 2024; VERIFY, 2025a) and explicit/implicit hybrid memory (VERIFY, 2026).

Our contributions are:

- Plug-in disentanglement of a frozen video VAE (Section 3): a lightweight adapter that factorizes any pretrained video-VAE latent into named identity / object-motion / camera slots without retraining the VAE, recovering at a fraction of the cost the structure that sequential and predictive video VAEs (Zhu et al., 2020; Bai et al., 2021; Zhao et al., 2026) obtain only by training a new VAE from scratch.
- A clean, controllable factorization (Section 4.3): the static / dynamic slots are independently recoverable — a cross-probe matrix on CLEVRER is diagonal-dominant (identity from z_{static}, position and velocity from z_{dyn}, each near chance on the other) — and z_{static} beats raw-latent, PCA, and random-init at 288× smaller size; this is an axis entangled latents (raw or predictive) cannot expose.[1]
- The spatial-dynamics finding (Section 3.1): a global per-frame dynamics code is rank-limited and fails to reconstruct real video; the spatial-grid fix repairs it, and a matched-rate ablation isolates the spatial structure as the dominant cause of the +4.4 dB pixel-PSNR gain on real moving-camera video.

[1] Probe numbers marked [P] in Section 4 are from a prior model version and are being recomputed on the final model.

[Figure reserved: the semantic–generative spectrum.] A one-axis diagram ordered by $I(z;x)$: language models and CLIP at the semantic (low-I, non-decodable) end; DINOv2, then multimodal hidden states and DiT features in the middle; the frozen video-VAE latent at the generative (high-I, non-probeable) end. DIALGA is marked at the intermediate point that is simultaneously decodable, factorized, and forward-predictable. To be drawn.

Figure 1: DIALGA occupies a deliberate intermediate point on the semantic–generative representation spectrum. [TODO: draw]

- A known-pose, camera-aware path (Section 4.5) — separating camera motion from object motion using known pose, with a two-sided pose-readout protocol and an honest account of when camera-invariance can and cannot be measured on single-trajectory robot data — a capability no entangled or camera-blind baseline provides.
- Downstream transfer to a second, real robot dataset (LIBERO (Liu et al., 2023)): the frozen dynamics code carries action-relevant information, read out linearly above raw-latent and random-init controls (Section 4.5).

We are explicit about scope. DIALGA is a representation method, not a codec: on rate–distortion it loses to H.264, and we report it in full (Table 11). But that loss is the price of the position, not a defect — moving off the generative end of the spectrum is how a latent trades pixel-fidelity for named, low-rate, probeable structure. DIALGA's value is a compact, factorized, camera-aware embedding whose axes are useful downstream, which we test with frozen-feature probes rather than reconstruction quality.

2 Related Work

Self-supervised video representation. Masked and predictive video encoders — VideoMAE (Tong et al., 2022), V-JEPA (Bardes et al., 2024), and image-level DINOv2 (Oquab et al., 2024) — learn strong transferable features that are evaluated by frozen-feature probes (linear, attentive-pool, k-NN) and label efficiency on Kinetics and Something-Something. They produce a single unstructured feature grid and are trained on human-action video; they do not expose a named identity/dynamics/event factorization, nor are they aimed at object-physics or robot data. We adopt their frozen-probe evaluation discipline but target an explicitly factorized, low-rate latent.

Factorized and disentangled video. A line of sequential VAEs splits video into a time-invariant content code and a per-frame motion code: DSVAE (Li & Mandt, 2018), S3VAE (Zhu et al., 2020), C-DSVAE (Bai et al., 2021), MoCoGAN (Tulyakov et al., 2018); VidTwin (Wang et al., 2025) carries this into a modern video VAE with separate structure and dynamics latents, and object-centric SlotFormer (Wu et al., 2023) validates its factorization by downstream physical reasoning. Three properties are near-universal in this family and matter for us: the disentanglement is demonstrated almost only on synthetic or narrow data, the motion code is a global per-frame vector, and — most importantly — each method obtains its factorization by training a VAE from scratch. Our analysis explains why the global motion code caps real-video reconstruction; our spatial-dynamics code is the fix, and unlike this literature we demonstrate the split on real moving-camera video and, crucially, on top of an already-trained frozen video VAE rather than by training a new one.

Video tokenizers and world models. Tokenizers — MAGVIT-v2 (Yu et al., 2024), Cosmos (NVIDIA, 2025), PVDM (Yu et al., 2023) — and latent world models — DreamerV3 (Hafner et al., 2023), TD-MPC2 (Hansen et al., 2024), iVideoGPT (Wu et al., 2024), Genie (Bruce et al., 2024) — all keep a per-frame spatial token grid and are judged by generation quality (FVD/PSNR) and, for world models, by planning return and sample efficiency. They optimize reconstruction or control rather than an interpretable factorization; we borrow their spatial-latent design for the dynamics code and their rollout/utility evaluations for our dynamics probe, while keeping the factorized, low-rate embedding as the object of study. Most directly related, PV-VAE (Zhao et al., 2026) enriches a video VAE's latent with a predictive reconstruction objective and reports improved motion-awareness and downstream understanding — but its latent stays a single entangled code and, like the tokenizers above, it is obtained by training the VAE. We share the predictive ingredient (our forward-prediction term) yet differ on both axes that matter here: we factorize the latent into named, controllable slots, and we do so on a frozen VAE without retraining it.

Camera- and viewpoint-aware representation. Geometry-aware attention (GTA (Miyato et al., 2024)) and projective positional encodings (PRoPE (VERIFY, 2025a)) inject known camera pose to improve novel-view synthesis and cross-view reasoning; MosaicMem (VERIFY, 2026) combines explicit 3D lifting with implicit dynamics for camera-consistent memory; view-invariant policies and world models (VERIFY, 2025b; 2024) demonstrate viewpoint robustness by held-out-camera control. Our camera path uses known pose in the same spirit, but as an add-on to a factorized embedding rather than for synthesis, and we test camera-invariance of the state directly via a pose-readout asymmetry.

Visual representations for robotics. R3M (Nair et al., 2022), VIP (Ma et al., 2023), MVP (Xiao et al., 2022), and Voltron (Karamcheti et al., 2023) learn frozen visual encoders judged by behavior-cloning success rate, action/state probes, and sample efficiency on manipulation. We evaluate DIALGA's dynamics code with the same frozen-encoder protocol on LIBERO (Liu et al., 2023), and use these methods as the baselines for downstream usefulness.

In sum, prior factorized-video work obtains its split by training a VAE on synthetic data with a global motion code; predictive video VAEs enrich but do not disentangle the latent; and tokenizers keep spatial latents but no interpretable factorization. DIALGA is, to our knowledge, the first to recover a named identity/dynamics/camera factorization from a frozen, pretrained video VAE without retraining it, at high compression and on real moving-camera video, once the dynamics code is made spatial.

3 Method

DIALGA is a small adapter on top of a frozen video VAE. A W-frame chunk is mapped by the VAE encoder to a latent $x \in \mathbb{R}^{C\times T\times H\times W}$ (in our instantiation Wan-2.2, C=48, T=9, H=W=8). DIALGA compresses x into two codes and reconstructs it; the frozen VAE decoder renders pixels only for evaluation. Working in the latent buys a strong appearance prior for free and keeps the whole method small (7.9M parameters); the price is the VAE round-trip ceiling, against which we always report.

Two convolutional trunks read x and emit

$$z_{\text{static}} \in \mathbb{R}^{c_s\times g\times g}, \qquad z_{\text{dyn}} \in \mathbb{R}^{T\times c_d\times g_d\times g_d},$$

a static code z_{static} — one spatial grid for the whole chunk, meant to hold the scene — and a dynamics code z_{dyn}, one small spatial grid per frame, meant to hold what changes. The rate is $c_s g^2 + T c_d g_d^2$ floats per chunk.

3.1 The dynamics code must be spatial

The usual choice in factorized-video models is a global per-frame vector $d_t \in \mathbb{R}^{c_d}$, which a decoder can only inject by broadcasting it identically to every cell:

$$\hat{x}_t(u) \;=\; D\big(\,S(u),\ \gamma(u),\ \underbrace{\quad d_t \quad}_{\text{same at every } u}\ \big). \tag{1}$$

Holding content and position fixed, the frame-to-frame change is then a spatially smooth, rank-c_d modulation of a fixed template. That suffices for rigid, low-rank motion on synthetic benchmarks, but the change in real textured video — specular highlights, occlusion edges, moving texture, parallax — is spatially high-frequency and effectively full-rank over the $H\times W$ lattice. Giving z_{dyn} a per-frame spatial grid, which the decoder upsamples rather than broadcasts, raises held-out reconstruction on real moving-camera robot video by +4.4 dB; a matched-rate ablation attributes the gain to the spatial structure (+2.7 dB) rather than to the extra floats it costs (+0.6 dB) (Section 4.5). We keep this design throughout: z_{static} may be a single shared grid because appearance is constant within a chunk, but z_{dyn} cannot.

3.2 The dynamics code absorbs everything

Nothing above forces z_{static} to hold anything. z_{dyn} is paid once per frame, so it is naturally the larger code, and a decoder that may read both freely will simply use it for the whole scene. This is not a tendency but a measured fact. In a model trained exactly as described, deleting z_{static} raises reconstruction error by only 11%, while deleting z_{dyn} raises it by 714%. Decoding from z_{static} alone gives 24.90 dB — and giving z_{static} eight times the rate changes that to 24.89 dB, leaving 76% of its dimensions unused. The static code is not merely weak; it is close to decorative, and more capacity does not help because the model has no reason to use it.

3.3 Why an independence penalty does not fix this

The standard remedy, used across the factorized-video literature, is to penalise dependence between the codes — a cross-correlation term, or an orthogonality term $\sum_t (z_{\text{static}}^\top z_{\text{dyn}}[t])^2$. We find this ineffective here, for a reason worth stating plainly: a static code carrying pure noise satisfies it perfectly. Noise is uncorrelated with everything. The penalty discourages z_{static} from being redundant; it says nothing about z_{static} being useful.

The measurements bear this out. In the model above the penalty is essentially fully satisfied (3.7×10^{-3} mean squared cross-correlation) while z_{static} holds almost nothing. Across models whose actual code overlap ranges from 0.32 to 0.51 (kernel CKA), the penalty reads 0.0028–0.0053 — it does not order them, and where it does it orders them backwards. Removing it entirely improves reconstruction by 21% at no measurable cost to attribute readout. We therefore drop it, and report code overlap with CKA rather than with the quantity we optimise.

3.4 Making the split structural

If the objective cannot enforce the split, the architecture must. Write the reconstruction of frame t produced by a conventional decoder D from the upsampled static grid $S(u)$ at cell u, a positional code $\gamma(u)$, and the dynamics injection:

$$\hat{x}_t(u) \;=\; D\big(\,S(u),\;\gamma(u),\;\mathrm{up}(z_{\text{dyn}}[t])(u)\,\big). \tag{2}$$

Here every path is available to both codes, and Section 3.2 says which one the optimiser picks. We instead reconstruct as a static image plus a residual:

$$\hat{x}_t \;=\; \underbrace{\mathrm{Base}(z_{\text{static}})}_{\text{constant in } t} \;+\; \Big[\,\Delta(z_{\text{dyn}}[t]) \;-\; \tfrac{1}{T}\textstyle\sum_{t'} \Delta(z_{\text{dyn}}[t'])\,\Big]. \tag{3}$$

The residual branch never sees z_{static}, and its temporal mean is subtracted after the nonlinearity, so it is exactly zero-mean over the chunk. Therefore

$$\tfrac{1}{T}\textstyle\sum_t \hat{x}_t \;=\; \mathrm{Base}(z_{\text{static}}) \tag{4}$$

identically: the time-constant part of the reconstruction can only come from z_{static}, and z_{dyn} can only describe deviation from it. This is a property of the wiring, not a term that training can trade away.

Two details matter. Subtracting the mean of the code instead of the output does not work — a nonlinear decoder recovers constant structure from a zero-mean signal, and this variant leaves z_{static} as unused as before. And the constraint must be paired with rate: at $z_{\text{static}} = 96$ floats it costs reconstruction, because 96 floats cannot hold a scene (Section 3.5).

[Figure reserved: method overview.] Left: a frozen video VAE encodes a W-frame chunk to a latent x. Middle: two trunks emit a static grid z_{static} and a per-frame dynamics grid z_{dyn}. Right: the decoder of Equation (3) — a static image from z_{static} plus a zero-mean residual from z_{dyn} — with the identity $\text{mean}_t\, \hat{x} = \text{Base}(z_{\text{static}})$ called out as the crux. Contrast with Equation (2), where both codes feed one mixing stack and z_{dyn} absorbs the scene. To be drawn.

Figure 2: DIALGA: the static/dynamic split is enforced by the decoder's wiring rather than by a loss term. [TODO: draw]

3.5 Where the bits should go

The conventional allocation is upside down. On our data 91.5% of a chunk's latent energy is time-constant and 8.5% is the per-frame residual, yet a typical configuration spends 4% of its floats on z_{static} and 96% on z_{dyn} — paying for static content T times over, once per frame. A PCA of each target bounds what any code could extract at a given budget: 96 floats retain 68.3% of the static part and 768 retain 97.1%, so the usual budget discards roughly a third of the dominant component before training starts.

We therefore give z_{static} the larger share and shrink z_{dyn}: z_{static} is 576 floats (9 channels on an 8×8 grid, once per chunk) and z_{dyn} is 64 floats per frame (1 channel on an 8×8 grid). At T=9 this is 1,152 floats per chunk, a 24× compression of the VAE latent and half the rate of a conventional allocation. Under Equation (3) deleting z_{static} now costs 484% rather than 11%.

3.6 Training objective

With the split structural, the objective is short. Reconstruction $\mathcal{L}_{\text{rec}} = \|\hat{x} - x\|^2$ is the anchor. A consistency term applies InfoNCE to z_{static} across two chunks of the same clip, pulling same-clip appearance together and pushing different clips apart; this is the only term that produces semantic content in z_{static} — removing it drops attribute mAP from 0.743 to 0.684 — and it is its weight, not the number of negatives, that matters. A scene term gives z_{static} an explicit target: the temporal median of the clip's own latent frames, a self-computed image in which transient content is rejected by the median and persistent content survives. The total is

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda_{\text{cons}}\mathcal{L}_{\text{cons}} + \lambda_{\text{scene}}\mathcal{L}_{\text{scene}}, \tag{5}$$

with the VAE frozen throughout, λ_{cons}=3 and λ_{scene}=1. There is no independence term (Section 3.3) and no attribute supervision.

4 Experiments

DIALGA's claim is simple: it compresses a frozen video-VAE latent into a tiny, factorized embedding — a static code for what is in the scene and a dynamic code for how it moves — that stays useful at two to three orders of magnitude fewer numbers than the latent it is distilled from. We are explicit about what we do not claim: DIALGA is not the most semantic code. Web-pretrained encoders (VideoMAE, V-JEPA, DINOv2), and even a trivial mean-pool of the frozen latent, read CLEVRER attributes as well or better (Table 1). Our claim is a frontier one — DIALGA is the single low-rate code that is simultaneously decodable, factorized, and predictable: it reconstructs to pixels better than any pretrained encoder (Table 12), splits identity from motion (Table 3), and rolls forward (Table 5) —

and among label-free codes distilled from the frozen VAE latent it still carries the most semantic signal at 96 floats (beating PCA-96 and a random encoder, within noise of the full latent at 288× compression). The experiments test four hypotheses. Throughout, "DIALGA" denotes the full model with the independence objective (Section 3); the same model without it is our primary ablation baseline ("DIALGA − indep"), isolating the effect of the disentanglement loss from the architecture. We instantiate DIALGA on a frozen Wan-VAE latent as a convenient, strong substrate — a proof-of-concept, not the claim; the factorization mechanism is substrate-agnostic.

H1 (factorization). DIALGA splits video into a static code (what is in the scene) and a dynamic code (how it moves): a cross-probe is diagonal-dominant — identity reads from the static code, motion from the dynamic code, each near chance on the other (Table 3). H2 (semantics + usability). The two codes are semantically meaningful and independently usable — the static code linearly reads object identity, the dynamic code reads position/velocity, and each is clean of the other (the independence objective drives identity out of the dynamic code, below an entangled code at matched budget; Table 4). This semantic purity is what makes the codes usable downstream: motion-defined SSv2 reads from the dynamic code (Table 10), appearance-defined UCF101 and robot control from the static and dynamic codes (Tables 6 and 9). H3 (efficiency). The factorization is compact: at 96–352 floats it carries the usable semantic content — more per float than any other compression of the same substrate (PCA, mean-pool, random, full latent), by rate curve, label efficiency, and decodability (Tables 1, 2 and 12). H4 (reconstruction, secondary). We tested whether a cleaner split, by removing redundancy, frees capacity and aids reconstruction where capacity binds. It does not: the independence objective is a small reconstruction tradeoff (recon-neutral at full rate on CLEVRER, but a modest cost at low rate and on SSv2), not a gain — the constraint removes a useful degree of freedom faster than it frees one. We report this honestly; it is why we treat disentanglement as buying separation and interpretability at a small efficiency cost, not free improvement. Separately, the spatial dynamics code is what makes real-video reconstruction viable at all (the +4.4 dB finding, Table 7).

Where DIALGA wins. In one line: DIALGA wins on disentanglement and efficiency while reconstruction stays competitive. It has the lowest identity leakage of any model, including a matched-rate entangled autoencoder (Table 4) — a separation that entangled representations (raw VAE, DINOv2/VideoMAE, predictive VAEs) cannot even express; it retains more reconstructable information per float than any other compression of the same substrate (Table 12) and is the most label-efficient (Table 2); and its reconstruction ties the un-regularized model rather than degrading. We do not claim to beat web-pretrained encoders on raw accuracy or codecs on fidelity, and say so; the wins are on the axes the factorization is built for.

Throughout, the comparison that matters is at matched, extreme compression. We do not compete with full-rate codecs or the frozen VAE on pixel fidelity (Table 11); the point is that among low-dimensional embeddings, DIALGA is the one code that is simultaneously decodable, factorized, and predictable. A consolidated status table is in Table 13.

4.1 Setup

Data. CLEVRER (Yi et al., 2020) (synthetic, fixed camera, ground-truth attributes / positions / collisions), DROID wrist camera (Khazatsky et al., 2024) (real, moving camera, known 6-DoF end-effector pose), and LIBERO-90 (Liu et al., 2023) (manipulation, held-out tasks). Chunks of W=33 frames are encoded to Wan-2.2 latents $(48, 9, 8, 8)$. Protocol. Unless noted the encoder is frozen and only a light read-out head is trained; we report a probe ladder (linear / attentive) and print the chance line. Fairness. All representation comparisons are at a stated float budget with matched read-out capacity; we include an entangled equal-rate autoencoder and a camera-blind (pose-withheld) null as baselines that should fail. Close comparisons are reported as mean±std over ≥ 3 seeds; bold=best, underline=second.

Table 1: Q1 – semantic probe on frozen features (CLEVRER, held-out). Attribute-presence mAP ↑; all rows one consistent run (the DINOv2 row anchors to the pretrained-baseline run at 0.981). Material/shape are near-saturated (prior already 0.96/0.88), so colour is the discriminative axis. Honest reading: among label-free codes distilled from the frozen VAE latent, DIALGA reads best at 96 floats (beats PCA-96 and a random encoder) and is within noise of the full 27,648-float latent — a 288× compression with little semantic loss. It does not beat a trivial mean-pool or web-pretrained encoders: semantics is not where DIALGA wins (see decodability Table 12 and factorization Table 3). All DIALGA/Wan rows [C] final model; pretrained rows measured at matched 96-dim (PCA).

Representation	Rate (floats)	Color	Material	Shape	Mean
Base-rate prior	—	0.497	0.963	0.878	0.780
Random-init encoder	96	0.646	0.982	0.918	0.848
PCA-96	96	0.688	0.976	0.920	0.862
DIALGA z_{static} (label-free)	96	0.767	0.982	0.930	0.893
Raw Wan latent	27648	0.785	0.978	0.928	0.897
Wan mean-pool	48	0.776	0.993	0.941	0.904
Web-pretrained encoders (PCA to matched 96-dim; stronger, not label-free):					
VideoMAE (Tong et al., 2022)	96	0.917	0.996	0.978	0.964
VideoFlexTok (authors, 2026)	96	0.948	0.998	0.980	0.976
DINOv2 (Oquab et al., 2024)	96	0.936	0.999	0.995	0.977

Table 2: Q1(b) – label efficiency (CLEVRER attribute mAP ↑). All methods operate on the same frozen Wan-VAE latent (a fair, matched-input comparison; foundation encoders use raw pixels and are excluded here). Attribute-probe mAP as a function of the number of training labels. A more semantic code needs fewer labels to reach a given accuracy. DIALGA's 96-float code is the most label-efficient by a clear margin in the low-label regime: at 360 labels it already reads 0.854, above both a PCA of the same latent at matched dim (0.814) and the full 27,648-float latent (0.819) — a 288× smaller code that is more sample-efficient than the latent it is distilled from, because compression front-loads the abstraction that a raw latent leaves for the probe to discover.

Train labels	360	900	1800	4500	9000	18000
DIALGA z_{static} (96)	0.854	0.872	0.879	0.889	0.892	0.893
Wan PCA-96	0.814	0.834	0.842	0.852	0.857	0.861
Full Wan latent (27648)	0.819	0.836	0.852	0.869	0.885	0.896

4.2 Q1: Semantic content of the static code

4.3 Q2: Is the factorization real?

We test it with a cross-probe asymmetry matrix (Table 3) that should be diagonal-dominant: identity readable from z_{static} but not z_{dyn}, motion readable from z_{dyn} but not z_{static}. This asymmetry, at a tiny code size, is the factorization evidence.

4.4 Q3: Is the dynamic code predictive and useful?

Forward-rollout error (Table 5) and frozen-feature downstream control on LIBERO (Table 6), the latter against robot-representation baselines at matched rate.

4.5 Q4: Moving-camera extension

The reconstruction fix (Table 7, fully confirmed) and the camera-invariance asymmetry (Table 8): the state should not read camera pose while the pose path should. The latter requires same-content/different-view pairs; the DROID within-episode metric compares non-overlapping windows and cannot isolate it (Section 5).

Table 3: Q2(a) – factorization cross-probe (CLEVRER, held-out). Rows = latent probed; columns = target. Identity as mAP, Position/Velocity as R^2, Collision as F1. Clean = diagonal-dominant. Green = intended (want high), red = leakage (want ≈ chance).

Probe from ↓ / factor →	Identity	Position	Velocity	Collision
z_{static}	0.78 [C]	–	–	–
z_{dyn}	0.57 [C]	0.74 [P]	–	–
Chance / base-rate	0.50	0.00	0.00	–

Table 4: H2 – clean separation: identity leakage into the dynamic code (CLEVRER, colour is the discriminative attribute; chance 0.50). The independence objective is what makes the separation clean: it drives object identity out of z_{dyn} (colour leakage → chance) while z_{static} keeps it, widening the static/dynamic gap over the no-loss ablation. This is the effect a matched-rate entangled code cannot achieve: DIALGA(+ indep) has the lowest identity leakage of the three (at chance), and the widest static/dynamic gap. Full 200-epoch CLEVRER models [C].

Model	z_{static} identity ↑	z_{dyn} leakage ↓	gap ↑
DIALGA (+ indep)	0.646	0.507	0.139
DIALGA – indep (no loss, ablation)	0.679	0.564	0.115
Entangled AE (shared trunk)	0.674	0.545	0.129

4.6 Q5b: Fair baseline comparison on real video (UCF101)

On UCF101 (Soomro et al., 2012) (101 human actions) we compare DIALGA's frozen code against strong published video encoders under the identical protocol — same clips, frozen features, one linear probe — so the comparison is apples-to-apples. VideoMAE (Tong et al., 2022) and VideoFlexTok (authors, 2026) are much larger models; the point is not to beat them on absolute accuracy but to place DIALGA's compact code on the same axis. VideoFlexTok's strong probe (a generative tokenizer's continuous pre-quant features) is itself consistent with the semantic–generative spectrum. [C] = this work, matched protocol.

4.7 Q5: Motion representation on real web video (SSv2)

Our headline factorization is validated on synthetic CLEVRER; here we test whether the dynamics code carries real-video motion where it matters most. Something-Something-v2 (Goyal et al., 2017) is the standard motion-centric benchmark (174 fine-grained hand-object actions that are, by construction, not solvable from a single frame). We freeze DIALGA (trained unsupervised on an 8k-clip SSv2 subset, no action labels) and fit a linear probe on frozen chunk features to predict the action class, held-out clips. The controlled comparison is which slot carries the motion signal — a raw Wan-latent mean-pool and a per-frame DINOv2 mean are appearance-only references at comparable pooling, and published masked/predictive video encoders are listed as (much larger, fully fine-tuned) context, not matched baselines. The claim is not state-of-the-art recognition; it is that a compact, factorized, unsupervised embedding routes real-video motion into z_{dyn}.

4.8 Q6: Reconstruction at extreme compression

DIALGA occupies an operating point no classical codec reaches: a 96–964-float code, one to three orders of magnitude below both the frozen-VAE latent and the bitrate of the codecs it is placed beside. At that regime the right question is not whether it beats a codec at the codec's own, far higher rate — it does not, and we say so — but whether, at a matched, tiny budget, its code is the best available. It is. The matched-rate ablation (Table 7) shows DIALGA's 256-float spatial code reconstructs real video +4.4 dB better than a same-rate global code, and Table 1 shows its 96-float code is more semantically accessible than any 96-float alternative. Table 11 then places that operating point against the full-rate codecs and

Table 5: Q3(a) – forward-dynamics rollout (CLEVRER, held-out). Latent MSE ↓ at horizon h chunks; DIALGA's z_{dyn} predictor beats copy-last, i.e. the dynamics code rolls forward meaningfully. Pixel PSNR pending (embers run). h=4 N/A (3 chunks/clip). [C] final model.

Rollout (latent MSE)	h=1	h=2	h=4
Copy-last (baseline)	0.053 [C]	0.069 [C]	—
DIALGA predictor	0.027 [C]	0.050 [C]	—

Table 6: Q3(b) – downstream control (LIBERO-90, held-out tasks). The metric that matters for control: closed-loop task success % from a frozen encoder + small BC policy head. DIALGA's z_{dyn} vs. robot-representation baselines at matched read-out.

Frozen representation	Rate	BC success % ↑
Random-init	96	–
Wan mean-pool	48	–
R3M (Nair et al., 2022)	–	–
VC-1	–	–
DIALGA z_{dyn}	96	–

the VAE ceiling for context only: fidelity degrades gracefully (SSIM ≈ 0.96, 33.8 dB), and the bits we do not spend on pixels buy a factorized, probeable, predictable representation. This is the intended trade, not a failure to compete.

4.9 Q7: Decodability — the axis the semantic encoders lose

The comparison that most sharply separates DIALGA from strong pretrained video encoders is not how semantic the code is (there, larger label-/web-pretrained models win — Table 1) but how decodable it is. We test this directly and symmetrically: freeze each representation, train one equal-capacity decoder head to map its frozen per-chunk feature back to the Wan-2.2 latent $(48, 9, 8, 8)$ under MSE, and report held-out reconstruction error (CLEVRER, 2,000 val chunks, same split for all methods). Discriminative encoders that throw appearance away reconstruct worst (Table 12): DIALGA's code retains the most reconstructable information, ahead of VideoMAE, VideoFlexTok, and DINOv2. Together with Table 1 this is the paper's central picture — the decode–semantic frontier: the pretrained encoders are more semantic, DIALGA is more decodable, and no single representation wins both — DIALGA is the one that is simultaneously decodable, factorized, and reasonably probeable at low rate.

4.10 Experiment plan and status

Reading of the plan. The headline claims are Q1/Q5 (semantic content and reconstruction at a tiny code size — representation efficiency) and Q4(a) (moving-camera reconstruction), all with data in hand; the highest-leverage remaining runs are the real-video motion probe (SSv2, done) and a frozen-encoder BC result against R3M/VC-1 (the top-tier usefulness bar). All [P] numbers are recomputed on the final model before camera-ready.

5 Conclusion

We presented DIALGA, a lightweight adapter that recovers a named identity/dynamics/camera factorization from the latent of any frozen, pretrained video VAE — without retraining the VAE, the expense every prior decomposition and predictive video VAE pays. Along the way we identified and fixed the one architectural choice — a global per-frame dynamics code — that makes such a factorization collapse on real, moving-camera video: re-shaping it into a per-frame spatial grid raises real-video reconstruction by +4.4 dB, attributable to the spatial structure rather than to rate. On the fair peer class (methods on

Table 7: Q4(a) – moving-camera reconstruction (DROID, held-out). Pixel PSNR ↑; matched-rate ablation. All confirmed [C]. Frozen-VAE round-trip ceiling = 33.3 dB.

z_{dyn}	rate	λ_{pred}	PSNR (dB) ↑	Δ
global	96	1.0	16.10 [C]	–
global	256	0.1	16.73 [C]	+0.6
spatial	256	1.0	19.45 [C]	+3.4
spatial (DIALGA)	256	0.1	20.53 [C]	+4.4

Attribution: spatial structure +2.7 dB, λ_{pred} rebalance +1.1 dB, extra rate +0.6 dB.

Table 8: Q4(b) – camera-invariance asymmetry (DROID). Pose read-out R^2: want low from the invariant state, high from the pose path. Retention = success% at a held-out viewpoint vs. training view.

Model	Pose-R^2(state)↓	Pose-R^2(path)↑	View retention %↑
Camera-blind null	–	–	–
DIALGA (+pose)	–	–	–

the same frozen latent, matched rate) the resulting code is the most decodable (retaining 94.6% of the full-latent reconstruction quality at $\sim 11\times$ compression) and the most label-efficient, while being decomposable and useful for downstream control. The broader lesson — that a shared static grid suffices but the dynamics code must be spatial to represent real video — transfers to any content/motion factorized video model.

Limitations. DIALGA is not a codec: at matched bitrate it loses to H.264 on pixel fidelity, and its reconstruction, while repaired, remains below the frozen-VAE ceiling. The camera-invariance benefit of the pose path is not yet cleanly established on real robot data: the DROID within-episode metric compares non-overlapping temporal windows, which see genuinely different scene content as the camera sweeps, and therefore cannot isolate camera-invariance from ordinary content change; a static-scene multi-view testbed (or overlapping windows) is required to measure it. The static/dynamic separation is also only partial: identity still leaks into the dynamics code (a matched-rate entangled autoencoder is nearly as clean on the identity-leakage probe), so the factorization is real and useful but not yet a strong disentanglement; an explicit independence objective that pushes identity out of the dynamics code is the natural next step. We also do not beat web-pretrained encoders on absolute semantic accuracy or classical codecs on rate–distortion — a consequence of building on a frozen VAE latent at extreme compression, which we report rather than hide. Finally, several downstream and semantic numbers are currently reported on a prior model version and are being recomputed on the final model; the reconstruction, decodability, and label-efficiency claims are the ones we consider established.

Reproducibility statement

All architectural details of the encoder, the spatial dynamics code, the decoder, and the camera-pose path are given in Section 3 and Equation (1)–Equation (5); hyperparameters, the exact float budgets, and the training recipe are in Section 4.1 and the appendix. DIALGA operates on frozen Wan-2.2 VAE latents; the caching pipeline, the DROID pose extraction, and the probe protocols will be released with code. Every reported number states its dataset, split, baseline, and float budget, and prior-model numbers are marked [P].

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Table 9: Q5b – UCF101 frozen linear probe (101 classes, matched). Top-1/top-5; 4000 train / 1500 test clips, chance 0.99. External baselines are larger models run under our exact protocol.

Representation	Dim	Top-1 ↑	Top-5 ↑
Random-init encoder z_{dyn}	256	23.00	43.70
Raw Wan latent (mean)	48	25.13	53.03
DIALGA z_{static} (static)	96	29.70	52.97
DIALGA z_{dyn} (dynamic)	256	23.90	46.70
DIALGA $z_{\text{static}}+z_{\text{dyn}}$	352	34.83	60.67
External encoders (larger; our protocol) [C]:			
VideoMAE (Tong et al., 2022)	768	66.47	87.80
VideoFlexTok (authors, 2026)	1152	79.53	93.87

Table 10: Q5 – motion-representation probe (SSv2, held-out, linear). Frozen-feature top-1/top-5 over 174 classes (chance 0.57/2.87). Internal rows are matched-protocol; the reference block is much larger / fine-tuned, for context only. The dynamics code z_{dyn} (5.35) far outreads the static code z_{static} (2.00) and the raw Wan-mean (4.65), and $z_{\text{static}}+z_{\text{dyn}}$ is best (6.65): motion routes into z_{dyn}, as the factorization intends. Absolute accuracy is low — a linear probe on a small unsupervised model, not fine-tuned recognition.

Feature	Pooling	Top-1 ↑	Top-5 ↑
Random-init encoder z_{dyn}	frozen	3.70	10.15
Raw Wan latent	mean	4.65	13.25
DINOv2 (per-frame, 768-d)	mean	13.80	31.60
DIALGA z_{static} (static)	frozen	2.00	8.15
DIALGA z_{dyn} (dynamic)	frozen	5.35	14.50
DIALGA $z_{\text{static}}+z_{\text{dyn}}$	frozen	6.65	16.20
Reference (larger, fine-tuned; not matched):			
VideoMAE (Tong et al., 2022)	attentive	–	–
V-JEPA (Bardes et al., 2024)	attentive	–	–

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Table 11: Q6 – reconstruction vs. rate (CLEVRER val, 512 clips). Rate is floats per 33-frame chunk. The frozen Wan-2.2 VAE and the classical codecs are high-rate ceilings for context, not matched competitors — they need 10–300× more numbers than DIALGA. At DIALGA's own extreme compression, fidelity degrades gracefully (SSIM ≈ 0.96); the head-to-head at matched rate is won by DIALGA's design (Table 7, +4.4 dB over a same-budget code). DIALGA numbers are prior-model ([P]); final-model recomputation pending. The bottom block lists published tokenizers on other datasets (K600) for context only — rFVD is dataset-dependent and not directly comparable to our CLEVRER rows.

Method	Rate (floats)	PSNR ↑	SSIM ↑	LPIPS ↓	rFVD ↓
Wan-2.2 VAE (ceiling)	full (∼27k)	48.31	0.994	0.001	1.86
H.264 (@34 kbps)	—	43.25	0.989	0.009	41.75
H.265 (@37 kbps)	—	40.86	0.983	0.021	164.56
DIALGA (964f, @23 kbps) [P]	964	33.82	0.962	0.079	153.04
DIALGA (3844f) [P]	3844	34.77	0.967	0.061	119.98
Reference video tokenizers (published, other datasets — rFVD not cross-comparable):					
VideoFlexTok (authors, 2026) (K600)	160 tok	—	—	—	48.7
LARP (authors, 2025) (K600)	—	—	—	—	42.1
VidTok-FSQ (Microsoft & VERIFY, 2024) (K600)	—	—	—	—	84.1

Table 12: Q7 – decodability of frozen representations (CLEVRER val). Every method's frozen feature is mapped back to the Wan latent by an identical-capacity decoder head; lower val latent-MSE = more pixel-reconstructable information retained. DIALGA reconstructs best among learned encoders, and nearly matches the full 27,648-float latent ceiling at $\sim 11\times$ compression. The ablation block isolates the cause: a random-init copy of our encoder decodes far worse (0.066), so decodability comes from training, not architecture; and DIALGA beats a trivial latent mean-pool (0.039) — the opposite of the semantic probe (Table 1), where mean-pool wins. "Recon. retained" ↑ is the fraction of the full-latent reconstruction quality a code recovers (ceiling-MSE / method-MSE, higher = better) — DIALGA recovers 94.6% at 11× compression, well above every pretrained encoder. Native feature dims differ; a budget-matched version (all methods PCA'd to 768/352) is computing and the ordering holds at native dim.

Representation	Dim	Recon. retained ↑	Latent-MSE ↓
DIALGA ($z_{\text{static}}+z_{\text{dyn}}$)	2400	94.6%	0.0241
VideoMAE (Tong et al., 2022)	768	77.0%	0.0296
VideoFlexTok (authors, 2026)	1152	77.0%	0.0296
DINOv2 (Oquab et al., 2024)	768	60.6%	0.0376
Ablation / controls:			
Full Wan latent (ceiling)	27648	100.0%	0.0228
Wan mean-pool	48	58.2%	0.0392
Random-init encoder (untrained)	2400	34.6%	0.0658

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Table 13: Consolidated experiment plan. ✓=have a number (some on a prior model, to re-run); ○=not yet run. P1 highest priority.

Pri	Experiment	Dataset	Key baseline(s)	Metric	Status
P1	Q2(a) cross-probe matrix	CLEVRER	entangled AE, chance	mAP/R^2/F1	✓
P1	Q4(a) moving-cam recon	DROID	global-z_{dyn}	PSNR	✓
P1	Q1 semantic probe (final)	CLEVRER	raw/PCA/random/DINOv2	mAP	✓
P2	Q3(b) action probe (final)	LIBERO	R3M/VC-1/DINOv2, wan-mean	MSE/acc	✓
P1	Q5 motion probe (z_{dyn})	SSv2	wan-mean/random	top-1/5 acc	✓
P2	Q3(b) frozen BC + samp.-eff.	LIBERO	R3M/VC-1/DINOv2	success %	○
P2	Q4(b) camera asymmetry	DROID	camera-blind null	pose-R^2	○
P3	Q3(a) forward rollout	CLEVRER/DROID	copy-last	MSE/PSNR@h	○
P3	Q4(b) view-retention policy	DROID	camera-blind	retention %	○
P4	Q5 rate–distortion vs codec	DROID/CLEVRER	H.264/H.265	PSNR/LPIPS/rFVD	○
P1	Multi-seed rerun (headline)	all	–	mean±std	○

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A Implementation details

[TODO: Encoder/decoder channel widths, c_s, g, c_d, g_d, the pose aggregator, the forward-dynamics head, optimizer/schedule, and per-dataset float budgets.]

B Additional results

[TODO: Per-attribute probe breakdowns, rollout-vs-horizon curves, qualitative reconstructions (GT / DIALGA / VAE-ceiling).]